Tianzhe Han

(612) 487-7273 han00510@umn.edu/hantz0120@proton.me

EDUCATION

University of Minnesota Twin Cities
Master of Science in Computer Science 09/2024-12/2026
GPA: 3.89/4.0

Wuhan University
Master of Engineering in Photogrammetry and Remote Sensing 2016
GPA: 3.81/4.0

Wuhan University
Bachelor of Engineering in Remote Sensing Science and Technology 2013
GPA: 3.66/4.0

SELECTED COURSE PROJECTS

Kriging-Informed Optimal Transport Conditional Flow Matching for Sea Level Anomaly Downscaling
CSCI 8715: Spatial Data Science Research, University of Minnesota Spring 2026

- Developed a kriging-informed Optimal Transport Conditional Flow Matching (Ki-OT-CFM) framework for probabilistic sea level anomaly downscaling.
- Integrated Universal Kriging conditioning, and variogram regularization to preserve spatial structure.
- Reduced training and inference costs compared with a diffusion-based baseline while maintaining competitive reconstruction quality.

Low-Resource Akkadian-to-English Translation with Byte-Level Transformers
CSCI 5541: Natural Language Processing, University of Minnesota Spring 2026

- Participated in a team project on low-resource Akkadian-to-English translation using ByT5, involving byte-level sequence modeling, hyperparameter tuning, inference optimization, and BLEU/chrF++ evaluation.
- Contributed to project brainstorming, research discussion, and preparation of the presentation and report.

Comparison of Latent Space Interpretability in Generative Models
CSCI 5527: Deep Learning, University of Minnesota Spring 2025

- Collaborated on a comparative study of latent-space interpretability across StyleGAN2, VAE, β-VAE, and β-TCVAE.
- Contributed to the motivation and analysis of semantic interpretability and controllable generation in latent spaces.
- Participated in quantitative and qualitative evaluation using linear-SVM-based semantic directions, classification metrics, and latent-space attribute editing.

Three-Dimensional Semantic Reconstruction of a Household Environment
CSCI 5561: Computer Vision, University of Minnesota Spring 2025

- Contributed to a 3D semantic mapping system integrating Sfm-based reconstruction, Mask R-CNN semantic segmentation, and multi-object tracking.
- Adapted and integrated an open-source SORT implementation for household-scene object tracking using Mask R-CNN detections, Kalman filtering, and IoU-based association.
- Evaluated the tracking module using manually labeled trajectories and MOT metrics, including IDF1, IDP, and IDR.

Transformer-Based Sea Surface Temperature Forecasting
CSCI 8523: AI for Earth, University of Minnesota Fall 2024

- Collaborated on a transformer-based framework for 1-10 days sea surface temperature forecasting using NOAA OISST data.
- Surveyed deep-learning approaches for spatiotemporal forecasting, including ConvLSTM, tranformers, graph neural networks, and Earthformer-style attention mechanisms.
- Contributed to research planning, model-design discussions, and analysis of forecasting approaches across different spatial and temporal settings.

RESEARCH EXPERIENCE

Master Thesis: Road Segmentation in Rural and Urban Areas

Wuhan University 2015-2016

- For the rural area that sharp gradient variation exists at the edges of roads and the road network structure is simple, Mean-shift segmentation is applied and the shape index is used to filter out the road area prior. A post-processing step, based on Graph Cuts (for grey image) or GrabCut (for RGB color image), built upon over-segments, is used to optimize the segmentation result.
- For the urban area that is complex in the scene, the disparity map is used to get the terrain mask of the original image. After conducting the foreground and background prior selection process based on connected component analysis and NDVI index, the GrabCut post-processing step built upon pixels is used in every window got along the road based on gradient calculation.

Project: Automatic updation and reconstruction of urban road netwoks from crowd-sourcing GPS trajectory data

Wuhan University 2014-2015

My work:

- Estimate the central location of crowd-sourcing GPS trajectory data based on Mean-shift clustering.

Project: The prior development of remote sensing application system in transportation from high-resolution remote sensing imageries

Wuhan University 2014-2015

My work:

- Generate road maps of highway in Xinjiang from high resolution satellite images by segmentation, thinning and rule-based line segment connection.
- Extract the feature vector of segmentation block, like SIFT, and feed them to SVM classifier to separate road area from other parts.

Undergraduate Thesis: Dome Extraction from Airborne LiDAR Point Cloud

Wuhan University 2013-2014

- Decompose the 3D point cloud into separated 2D arcs according to the scanline by D-P algorithm and detect each arc based on Hough Transformation, thus extracting the dome structure out from the filtered point cloud.

Project: 3D reconstruction of Natural Surface and man-made objects from High-resolution remote sensing imageries

Wuhan University 2013-2014

My work:

- Improve the tensor voting method for linear structure saliency computation to accelerate the process of road extraction from point cloud based on super-voxel segmentation.

INDUSTRY EXPERIENCE

Geovis Technology Co., Ltd. Wuhan, China

Algorithm Engineer May-July 2024

- Flood area extraction from multi-source remote sensing data (multi-spectral and SAR) using mmsegmentation
- Literature reading in change detection, including binary change detection and multi-category semantic change

detection

Part-time Study / Research / Industry Collaborations May-Dec 2023

- Multimodality and Vision Foundation Models and their follow-up works in segmentation, detection, etc. vision tasks.
- Diffusion Models and AIGC
- Mathematical Foundations of Reinforcement Learning & Hands-on Reinforcement Learning

Project: Thematic Element Extraction and Analysis for Potential Geological Disaster Prevention

- Analyze the changing trends of surface deformation cluster according to multi-phase remote sensing data.
- Analyze the contribution of different key elements for geological disaster prevention: Moran's I, PCA, CCA.

Hubei NineCube Microelectronics Co., Ltd. Wuhan, China

Algorithm Engineer Apr-Sep 2022

- Implement geometry-related algorithms in Python to assist the design of chips (EDA)
- Translate matlab-written algorithms into C++. These algorithms include Differential Evolutionary Algorithm, which aims to find the optimal combination of (w, s, n, r) that can maximize the electromagnetic simulation result; and a small part of fast multipole method (FMM), which aims to determine the number of remainder terms when given out the source and target group in FMM.
- Fitting function using MLP, in order to apply machine learning/deep learning methods in scalable model estimation.

Remote-Sensing Algorithm Project

Collaboration with Beijing Zhixin Remote Sensing and Geographic Information Technologies Inc.

Feb-Mar 2022

- Re-implemented remote-sensing algorithms in Python, including QAA, Ocean Color Index, Net Primary Production (VGPM) in Python

Black Sesame Technologies Inc. Shanghai, China

Algorithm Engineer Aug-Dec 2018

- Calibrate VLP-16 with stereo camera. Semi-automatically extract the corner points and feed them into EPnP solver to calculate the [R|t] parameters. Reproject the LiDAR points to the camera coordinate system using these parameters to calculate the reprojection error and assess the calibration result.
- Apply typical deep learning models in stereo matching task.

TEACHING EXPERIENCE

University of Minnesota Twin Cities

Teaching Assistant – CSCI 2011: Discrete Structures of Computer Science Spring 2026

Teaching Assistant – CSCI2021: Machine Architecture and Organization Fall 2025

Teaching Assistant – CSCI 4707: Practice of Database Systems Spring 2025

SKILLS

Programming: Python, C/C++, MATLAB

Deep Learning Framework: PyTorch

Github Link: https://github.com/hantianzhe

HONORS AND AWARDS

The Third Prize Scholarship for Freshman, Wuhan University 2008

Excellent Student & The Third Prize Scholarship, Wuhan University 2009

Excellent Student & The Second Prize Scholarship, Wuhan University 2010

The Third Prize for Excellent Team in Social Practice, Wuhan University 2009

CSC (China Scholarship Council) Scholarship 2016